SE 2226 Test Completion Report

SauceDemo and Library-DB-Express

Based on ISO/IEC/IEEE 29119-2 Section 7.4

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1. Summary of testing performed

Testing was performed for two applications: SauceDemo and Library-DB-Express. SauceDemo was tested as a black-box web application. Library-DB-Express was tested locally with unit, coverage, mutation, and load testing tools.

For SauceDemo, the team tested login, logout, cart actions, checkout, product sorting, product detail page, locked user login, empty input fields, and wrong credential cases. The main techniques were Equivalence Class Partitioning, Boundary Value Analysis, Decision Table Testing, and Use Case Testing. Selenium IDE was used for recorded GUI flows, Selenium WebDriver Java was used for 2 coded tests, and Playwright was used as extra cross-browser evidence.

For Library-DB-Express, Jest tests were executed against the actual Express routes. The tests covered book listing, search, create book, validation errors, book detail page, update book, delete book, and unknown routes. Jest coverage, Stryker mutation testing, and JMeter load testing were also completed.

Main constraints were time, tool setup, and the difference between the planned REST-style routes and the real form-based routes of the Library-DB-Express project.

Area	Tool / Evidence	Result
SauceDemo GUI testing	Selenium IDE	9 recorded flows were prepared and reviewed.
SauceDemo coded browser tests	Selenium WebDriver Java	2 tests passed: valid login and invalid login.
SauceDemo cross-browser tests	Playwright	63/63 executions passed on Chromium, Firefox, and WebKit.
Library unit testing	Jest + Supertest	48/48 tests passed.
Library coverage	Jest --coverage	Statement 97.39%, Branch 78.26%.
Library mutation testing	Stryker	105 mutants tested, mutation score 60.19%.
Library load testing	JMeter	10, 50, and 100 user load tests executed. Error rate 0.00%.

2. Deviations from planned testing

Some small deviations happened during execution:

- The test plan described Library-DB-Express like a REST API with GET, POST, PUT, and DELETE /books routes. During testing, the actual application worked more like a form-based Express web application. Therefore, tests were written for the real routes such as GET /books, POST /books/new, POST /books/:id, and POST /books/:id/delete.
- The JMeter .jmx file shows the last used configuration. However, the load test was run separately for 10, 50, and 100 users, and screenshots were saved for each result. This is a documentation/configuration difference, not a failed test.
- A separate defect tracking tool such as Jira was not used. Because no real application defect was found, a simple defect log table was written inside this report.
- Playwright was not the main planned tool, but it was used as additional evidence for cross-browser execution.

3. Test completion evaluation

Most planned completion criteria were met. The main test activities were completed and evidence files were produced. The only weak point is that the Library testing had to follow the actual Express routes instead of the originally written REST-style route names.

Completion criterion	Target / Plan	Evaluation
All planned test cases executed	At least once	Met for the available automated tests and recorded GUI flows.
SauceDemo WebDriver Java	At least 2 tests	Met. 2 Java WebDriver tests passed.
Compatibility testing	Firefox pass documented	Met. Playwright report shows Chromium, Firefox, and WebKit passed.
Library statement coverage	>= 70%	Met. Statement coverage was 97.39%.
Library branch coverage	>= 60%	Met. Branch coverage was 78.26%.
Mutation score	Reported	Met. Stryker mutation score was 60.19%.
JMeter average response time	< 500 ms at 50 users	Met. 50 users average was 20 ms.
JMeter error rate	< 5%	Met. Error rate was 0.00% for 10, 50, and 100 users.
Critical defects documented	All S1 defects documented	Met. No S1 defect was found.

Compatibility result

No functional compatibility problem was observed between Chromium and Firefox. All browser executions passed. There were only timing differences. Firefox was faster in some login/error tests, but Chromium was faster in some use-case flows such as checkout, product sorting, and navigation. These speed differences did not change the test results.

Example test	Chromium	Firefox	Comment
TC01 Valid Credentials	5.1 s	3.1 s	Firefox was faster.
TC06 Locked Out User	5.3 s	4.5 s	Firefox was slightly faster.
TC13 End-to-End Purchase	1.6 s	3.3 s	Chromium was faster.
TC16 Logout	2.3 s	3.5 s	Chromium was faster.
TC20 Product Detail + Add to Cart	1.5 s	2.8 s	Chromium was faster.
TC21 Cart Badge + Reset State	2.8 s	3.4 s	Chromium was faster.

4. Factors that blocked progress

The main problems were setup and small planning mismatches. They were solved during the testing process.

Blocking factor	Effect	Solution
Library route mismatch	The planned REST-style route names did not fully match the real application.	Tests were adapted to the actual Express form routes.
Tool setup	Jest, Stryker, Selenium drivers, and JMeter needed setup time.	Tools were installed locally and results were captured as screenshots/text reports.
Database state	Tests could affect each other if data was not reset.	Seed data was reset before tests.
JMeter configuration evidence	The final .jmx file only shows the last configuration.	Separate screenshots were used as evidence for 10, 50, and 100 users.
No defect tool	No Jira-style defect report existed.	A simple defect log table was added to this report.

5. Test measures

5.1 Overall measures

Measure	Result
SauceDemo Playwright test cases	21
SauceDemo Playwright executions	63 total: 21 Chromium + 21 Firefox + 21 WebKit

SauceDemo Playwright result	63 passed, 0 failed, 0 flaky, 0 skipped
SauceDemo Playwright total time	37.2 seconds
Selenium IDE flows	9 recorded flows
Selenium WebDriver Java tests	2 passed
Library Jest tests	48 passed, 0 failed
Jest test suites	1 passed, 1 total
JMeter load executions	3 load levels: 10, 50, 100 users
Defects found	0 application defects found in executed tests

5.2 Coverage measures

Coverage metric	Result	Target	Status
Statements	97.39%	>= 70%	Passed
Branches	78.26%	>= 60%	Passed
Functions	100%	Not fixed	Reported
Lines	97.34%	Not fixed	Reported

5.3 Mutation measures

Mutation metric	Result
Mutated files	2 source files
Total mutants	105
Killed mutants	43
Timed out mutants	19
Survived mutants	39
No coverage mutants	2
Errors	2
Mutation score	60.19%

5.4 JMeter load test measures

Users	Samples	Average	Min	Max	Std. Dev.	Error %	Throughput
10	10	23 ms	13 ms	60 ms	14.37	0.00%	11.9/sec
50	50	20 ms	10 ms	84 ms	18.61	0.00%	53.9/sec
100	100	237 ms	59 ms	428 ms	80.92	0.00%	82.5/sec

5.5 Simple defect log

No real product defect was found in the executed tests. Because of that, the defect log is kept very simple.

Defect ID	Application	Description	Severity	Status
D-001	SauceDemo	No functional defect found. All automated cross-browser tests passed.	None	Closed
D-002	Library-DB-Express	No critical functional defect found in Jest tests.	None	Closed
D-003	Library-DB-Express	No load/performance defect found. JMeter error rate was 0.00%.	None	Closed

6. Test deliverables

Deliverable	Location / File name
Test Plan	SE2226_Test_Plan.pdf

Test Completion Report	This document
SauceDemo Selenium IDE project	SauceDemo_Selenium_IDE.side
SauceDemo Selenium IDE evidence	Recorded execution videos and screenshot with 9 test names
SauceDemo Selenium WebDriver Java project	ValidLoginTest.java, InvalidLoginTest.java, pom.xml
SauceDemo WebDriver result evidence	Selenium_WebDriver_Java_Result.png
SauceDemo Playwright tests	SauceDemo_Playwright_Tests.spec.ts
SauceDemo Playwright report	SauceDemo-Playwright.pdf
Library Jest tests	books.test.js
Jest coverage report	Covarage.txt
Stryker configuration	stryker.config.json
Stryker mutation report	Mutation test.txt and HTML mutation report
JMeter test plan	Test Plan.jmx
JMeter load testing evidence	JmeterLoadTesting.docx

7. Lessons learned

- The test plan should be checked against the real application routes before writing all tests.
- Screenshots and tool output files are useful, because they make the final report easier to support.
- A small defect log should be prepared even when no defects are found.
- Cross-browser testing should compare functionality first. Speed differences are useful, but they are not defects by themselves.
- Mutation testing showed that passing unit tests are not always enough. Some survived mutants show where stronger assertions could be added later.
- For future work, we would keep the JMeter file with separate thread groups for 10, 50, and 100 users from the beginning.